

# Kayra Arai Ozturk

+1 604-771-1341 | [kayraarao@gmail.com](mailto:kayraarao@gmail.com) | [linkedin.com/in/kayra-arai-ozturk](https://linkedin.com/in/kayra-arai-ozturk) | [github.com/kiwaku](https://github.com/kiwaku) | [Portfolio](#)

## EXPERIENCE

### Enterprise Security – Intern

Summer 2024

*Mercari (TSE: 4385)*

*Tokyo, JP*

- Designed and shipped a Chrome extension enforcing OS- and role-aware **Slack workspace access control** (Declarative Net Request API, dynamic network-request rules, custom headers) **in-extension** rather than via a corporate proxy – removing the rollout dependency on VDI/fleet middleware and cutting per-user hardware spend **67%** (JPY 450K → 150K/yr) by enabling a shift from Mac to ChromeOS for managers.
- Built real-time **password-field detection** on unmanaged SaaS (Watcher class, async listeners, iframe/password hooks, whitelist/DB checks, Chrome message passing, Firebase-logged categorization), reducing data-breach exposure from unauthorized SaaS and partnering with security/legal on **BYOD** policy.

### Analytics & Data Science – Invited Collaborator

Oct. 2025 – Jan. 2026

*Clubhouse · Matsumoto Yamaga FC (J-League)*

*Tokyo, JP · Hybrid*

- Rebuilt fragmented semi-manual analytics into an automated **5-source** Python pipeline with unified athlete identity, deduplication, and **hierarchical** modeling – **34** Yamaga athletes in the direct workflow, **157** across the broader pipeline.
- First reproducible analytics pipeline in the group for a **J-League**-affiliated club; still in production after handoff, powering coach-facing physical, nutrition, and attendance reporting.
- Also administered the **Supabase** database, accounts, and general security across several university soccer teams and a baseball program.

### Co-Founder & CTO

Oct. 2022 – Present

*LAIVU – early-stage investor-backed startup*

*Tokyo, JP*

- Secured **\$35K+** in investor funding (early **\$20K** tranche) via founder pitches; earned broadcast coverage on **NHK**, Japan's national public broadcaster.
- Led product and engineering; owned compliant cross-border B2B operations, and now serve as sole full-stack engineer (**TypeScript**, **Firebase**, **React**) building **Fittomo**, a health-focused mobile app live on the App Store.

**AI Research Assistant** (*Mobiz, Jun. 2023 – Sep. 2023, Remote*): Built citation-backed PDF-QA and OCR pipelines with **LangChain**; foundation for later personal LLM systems work (2025–2026) spanning stateful agent orchestration, queue-backed recovery, and multi-provider safety evaluation.

## PROJECTS

### RedLLM – Adversarial LLM Safety Evaluation | *Python, DSPy, LiteLLM*

- Built a modular, **provider-agnostic** LLM-safety evaluation harness on **LiteLLM/DSPy** across **3 providers** (OpenAI, Anthropic, Together), with agent-style multi-variation attack generation, an ensemble judge (*AdaptiveJudge*), and token-level controls for reproducibility – the application-level AI tooling this role builds.

### Hephaestus – Local-LLM Serving & Secure Lab | *llama.cpp, SQLite, Proxmox VE, SSH*

- Served a private local model (**24 GB** VRAM, **64k–96k** ctx) over SSH tunnels with **SQLite durable queues**, Mode A/B failover, and a **10/10** automated acceptance suite – MLOps-style reliability for an AI service.
- Built the underlying **default-deny** cyber range on a dual-socket Xeon host with bastion-mediated NAT/DNS, GPU passthrough, and isolated attacker/victim/sniffer segments.

### Ferrum II – GPU IDS/IPS via RDMA | *C, CUDA, RDMA, libibverbs*

- Built a zero-copy GPU IDS/IPS that **RDMA**-writes packets from ConnectX-3 NICs directly into Tesla P40 VRAM, with GPU-side parsing (65,536-slot ring buffer) and GPU-generated TCP/ICMP responses for inline blocking.
- Sustained **2.62M pps (31.8 Gbps)** at **82  $\mu$ s** avg / **119  $\mu$ s** p99 latency – **50 $\times$**  lower than CPU baseline, **99.8%** classifier accuracy.

### RF & VEX Controller Pentest | *RTL-SDR, HackRF, Python*

Feb. 2022 – Jun. 2022

- Analyzed VEX v5 RF controller traffic and built replay/injection tooling that demonstrated remote motor-command hijacking without physical access; measured exploit reliability under range/noise, delivered mitigation guidance, and secured full project funding via live demo to IT stakeholders.

## EDUCATION

### The University of Manchester

Manchester, UK

*Bachelor of Science (Hons) in Computer Science*

*Aug. 2022 – June 2025*

## TECHNICAL SKILLS

**AI & LLM:** LiteLLM, DSPy, agentic/LLM orchestration, adversarial & safety evaluation, llama.cpp, PyTorch, TensorFlow

**Languages:** Python, TypeScript, JavaScript, C, C++, SQL, Bash, Assembly (ARM)

**Systems & DevOps:** Linux (Debian), **RDMA/libibverbs**, Git, Docker, systemd/launchd, SSH/ProxyJump, SQLite, Supabase, durable queues/failover, Proxmox VE, iptables/NAT, Wireshark

**Open Source:** EleutherAI / MLCommons – merged upstream contributions

**Certifications & Activities:** **CRTO** – Certified Red Team Operator (Zero-Point Security, 2026) | HeatWave White Hat Program | President, Manchester Japanese Society (2023–24, 500+ members) | **GreatUniHack 2024** (4th of 40+)